

CST2120

Web Applications and Databases

Coursework 1 – Game Website

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Project Overview

Purpose

TypeRush is a pirate-themed web-based typing game designed to improve typing speed and accuracy through a series of increasingly challenging levels. The game combines educational typing practice with interactive game mechanics such as scoring, lives, streak bonuses, rankings, and level progression. Players must complete different typing challenges while maintaining high accuracy and speed in order to achieve the highest possible score. The game was developed to make typing practice more engaging and enjoyable by introducing competition, rewards, and progression systems that motivate users to improve their performance.

The overall design of the game is inspired by the anime One Piece, with a pirate-themed interface, voyage-based progression, and nautical terminology used throughout the website. These elements help create a more immersive experience while maintaining the primary goal of developing typing skills in an entertaining and interactive way.

Technologies Used

The project was developed using HTML, CSS, and JavaScript. HTML was used to create the structure of the website, CSS was used to design the visual appearance and animations, and JavaScript was used to implement interactivity, game mechanics, user authentication, data storage, and score calculations. These technologies were combined to create a responsive and interactive web application that provides a complete gaming experience. Additional features such as form validation, browser-based data storage, dynamic content updates, and audio controls were also implemented to improve functionality and enhance the overall user experience.

Website Structure

Homepage

The homepage serves as the main entry point to the website. It introduces the TypeRush game, displays navigation links, and provides access to registration, login, settings, rankings, and the game itself.

Game Page

The game page contains the core gameplay functionality. Players can select levels, complete typing challenges, view live statistics, and receive result summaries at the end of each session.

How To Play Page

The How To Play page explains the rules of the game, describes each level, explains scoring mechanics, and provides screenshots to help new users understand the gameplay experience.

Profile Page

The profile page displays information related to the logged-in user, including personal details and game statistics such as highest score, best accuracy, best typing speed, and levels unlocked.

Terms and Conditions Page

This page outlines the rules governing the use of the website, user responsibilities, acceptable behaviour, and limitations of liability.

Privacy Policy Page

The Privacy Policy explains what information is stored by the website, how it is used, and how user data is handled.

Homepage

The homepage welcomes users to TypeRush and introduces the pirate-themed typing challenge. A navigation bar provides access to the various sections of the website, including Home, Game, How To Play, Rankings, Profile, and Settings.

The Start Voyage button allows users to begin playing the game. If a visitor is not logged in, they are encouraged to register or log in before accessing gameplay features. Logged-in users can immediately access the game.

The Settings menu allows users to adjust brightness and sound preferences. Registration and login are available through modal forms, while the Logout button becomes visible only when

a user is logged in. The Profile button provides access to the user's personal information and game statistics.

The homepage also serves as a central hub from which all major website features can be accessed. Its layout was designed to be simple and user-friendly, ensuring that both new and returning users can easily navigate the website, manage their accounts, view their progress, and quickly start playing the game.

In addition, the homepage provides users with an overview of the game's theme and objectives before they begin playing. Visual elements inspired by the pirate theme are used to create an engaging first impression, while clear navigation and organised content help users quickly locate important features and information throughout the website.

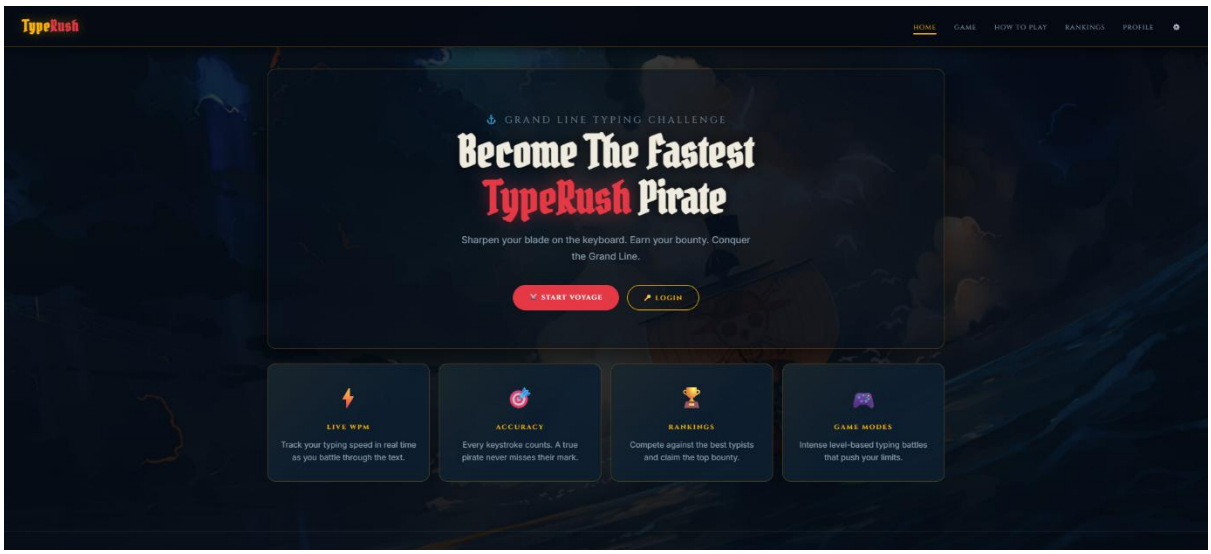


Figure 1: Homepage of TypeRush website

Registration and Login

The registration system allows new users to create an account by providing their first name, surname, age, email address, username, and password. Live validation is used to provide immediate feedback while users enter information. This helps prevent invalid data from being submitted and improves user experience.

During registration, usernames and email addresses are checked to ensure they are unique. Password confirmation is also required before an account can be created successfully.

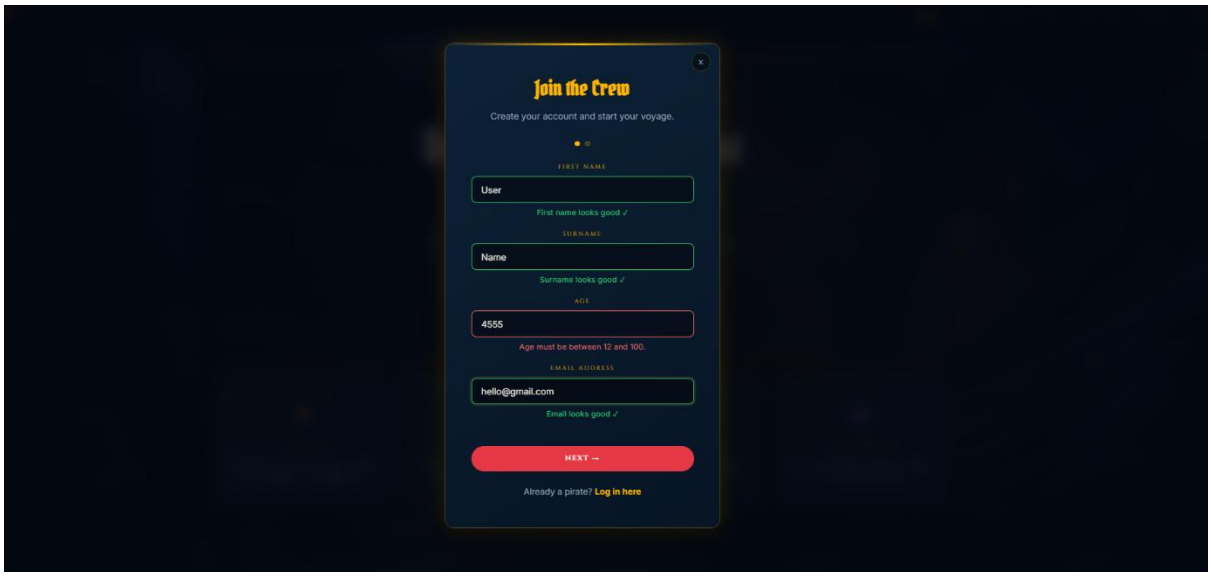


Figure 2: User registration form with live validation feedback

The login system allows existing users to authenticate using their username and password. Appropriate error messages are displayed if an account does not exist or if an incorrect password is entered.

Once logged in, the user session remains active throughout the website, allowing access to game progress, profile information, rankings, and unlocked levels. Password visibility controls are also provided to improve usability by allowing users to view or hide their passwords while typing. User account information is stored locally within the browser, enabling returning users to access their accounts and previously saved progress without needing to register again.

Additional validation checks are performed to ensure that all required fields are completed correctly before submission. Visual feedback is provided through colour changes and validation messages, helping users identify and correct errors quickly. This reduces the likelihood of incorrect information being entered and contributes to a smoother and more user-friendly registration and login process.

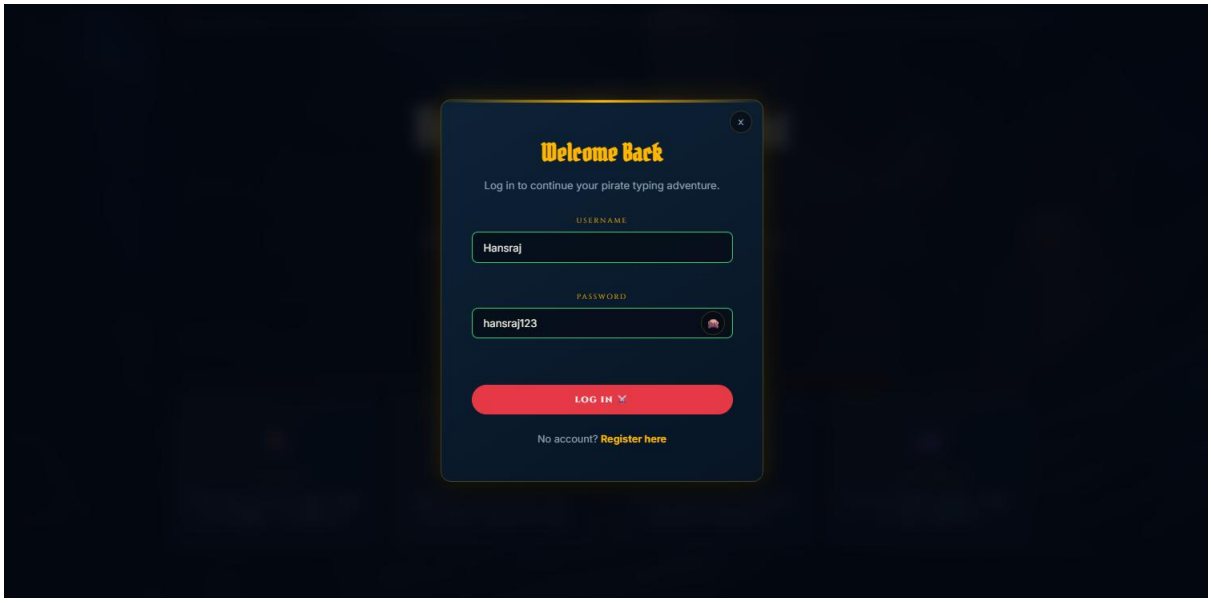


Figure 3: User login interface

Game Page

The game page contains three levels of increasing difficulty, each designed to test different typing skills.

Level 1 – Image Typing

In Level 1, players are presented with a random set of pirate-themed images and must correctly type the word represented by each image. This level focuses on recognition and spelling speed.

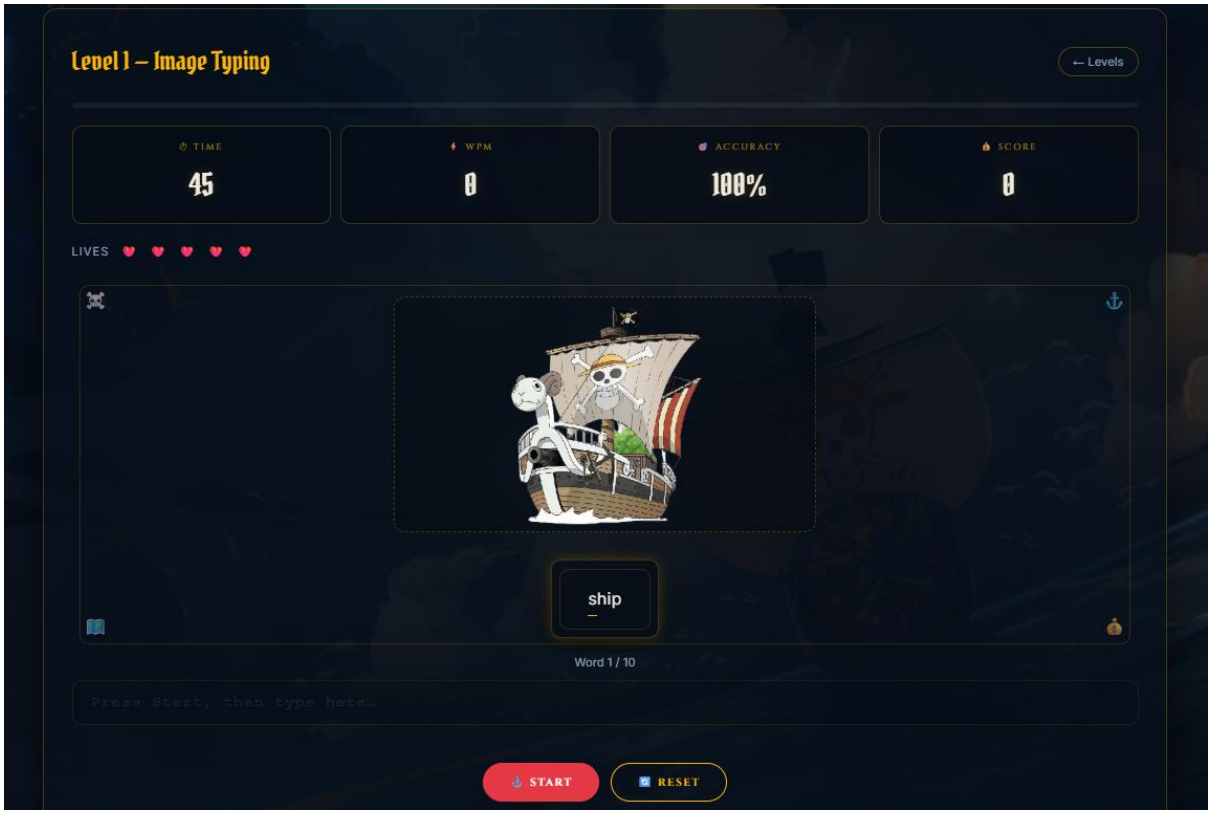


Figure 4: Level 1 image recognition challenge

Level 2 – Paragraph Race

Level 2 presents a paragraph which must be typed accurately and quickly. Live feedback highlights correct and incorrect characters, allowing users to monitor their progress in real time.

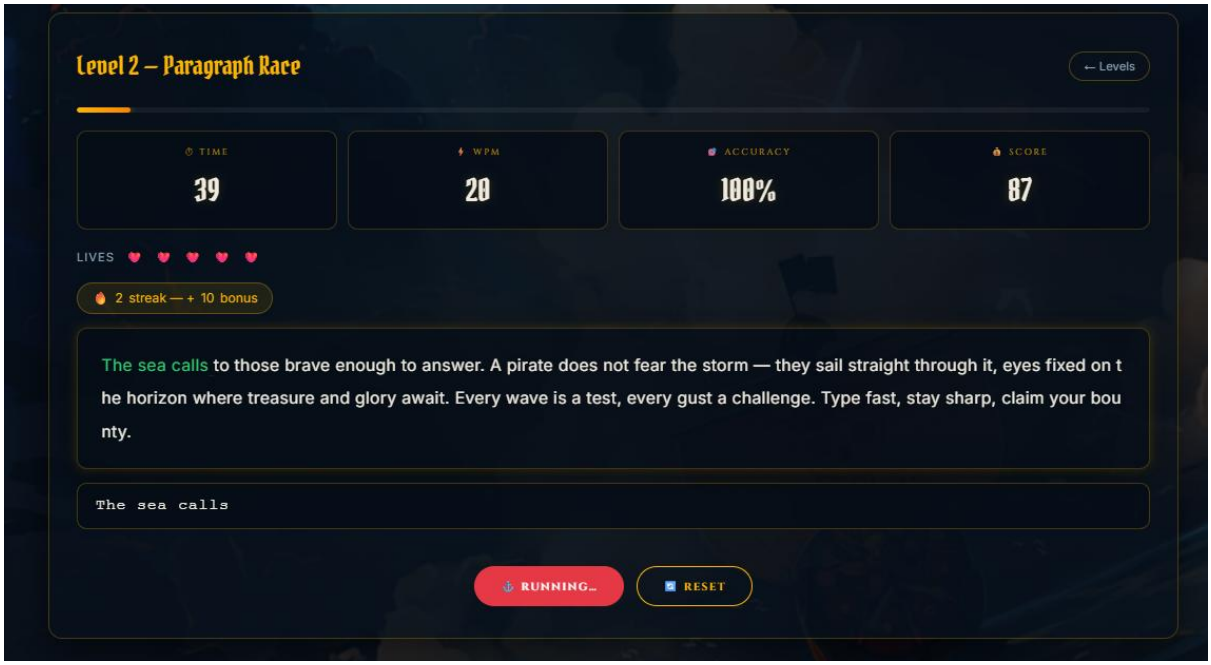


Figure 5: Level 2 paragraph typing challenge

Level 3 – Audio Dictation

In Level 3, players listen to an audio recording and type the words they hear. This level combines listening skills, memory, and typing accuracy.

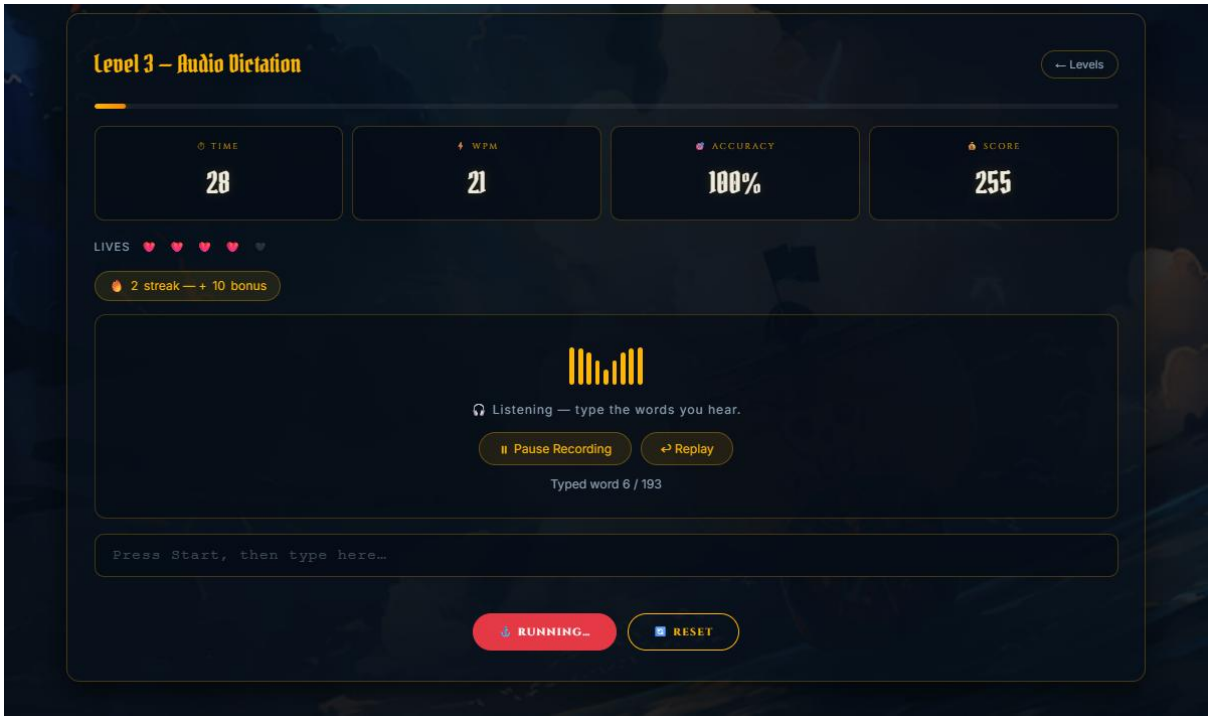


Figure 6: Level 3 audio dictation challenge

Scoring System

Scores are calculated using a combination of typing accuracy, typing speed, completed words, and streak bonuses. Consecutive correct answers increase the player's streak, resulting in additional bonus points. Mistakes reduce the score and also decrease the number of available lives.

Lives and Progression

Each level provides five lives. Losing all lives ends the session immediately. Levels are unlocked progressively based on successful performance, encouraging players to improve before advancing.

Results, Feedback and Progression

At the end of each level, a results screen is displayed showing the player's performance statistics, including score, words per minute (WPM), accuracy percentage, mistakes made, streak bonus earned, and time taken to complete the challenge. Based on the final score, the player is also awarded a rank which reflects their overall performance during the session.

Visual feedback is provided throughout the game using animations, graphical effects, and custom result screens. Successful completion displays a victory screen with themed graphics, while unsuccessful attempts display a different results screen indicating that the challenge was not completed successfully. These visual elements were included to make the game more engaging and rewarding for players.

Player progression is controlled through a level unlocking system. Access to higher levels depends on the player's performance and score achieved in the current level. Players who meet the required score threshold can unlock the next challenge, while those who do not achieve the required score must replay the level and improve their performance before progressing further.

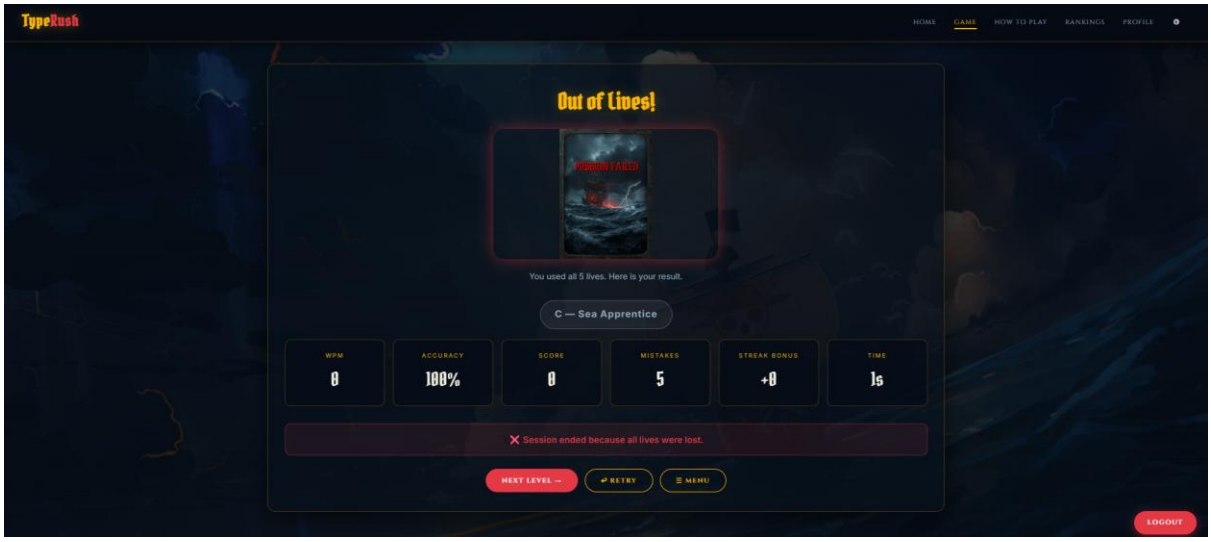


Figure 7: Results screen displayed after completing a level

Visual and Audio Feedback

The game provides live updates for score, WPM (Words Per Minute), accuracy, remaining lives, and progress. Audio recordings are used during Level 3, while animations and graphical effects enhance the user experience.

How To Play Page

The How To Play page serves as a user guide for the game. It contains an overview of TypeRush, explanations of the three levels, descriptions of the scoring system, and screenshots demonstrating important gameplay features.

The page is designed to help new players understand the rules before beginning the game and includes navigation options that allow users to return to the homepage or start playing immediately.

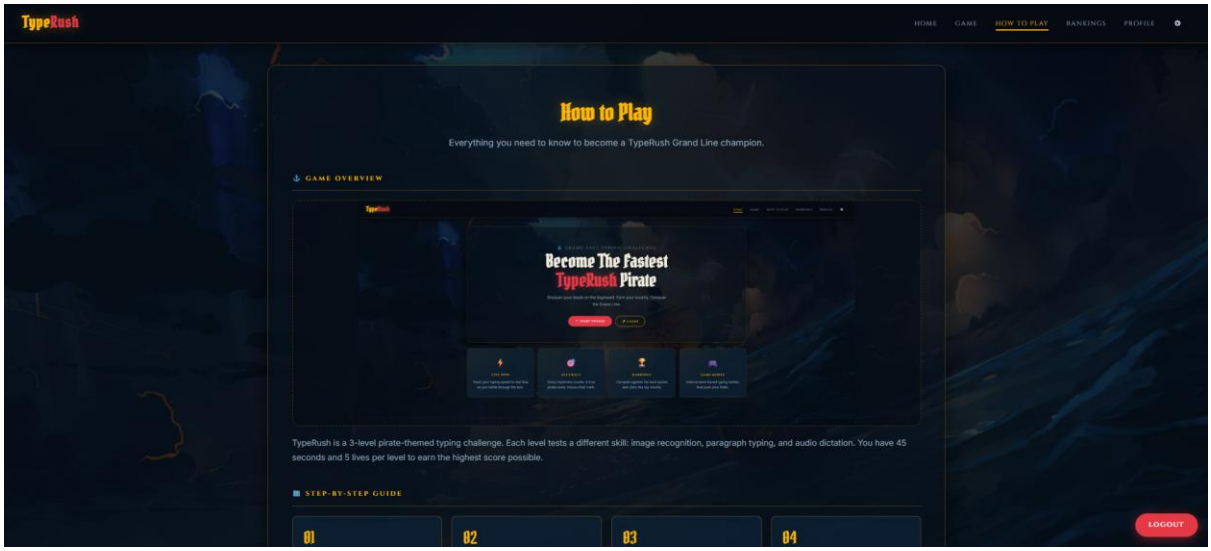


Figure 8: How to play page

Leaderboard

The leaderboard displays the highest-performing players based on their recorded scores. Rankings are updated whenever a user completes a game session and achieves a qualifying score.

To prevent duplicate entries, only the best score for each user is retained. The leaderboard provides a competitive element by allowing users to compare their performance with other players. Additional information such as rank, score, and player name is displayed to make comparisons easier and encourage competition between users.

Leaderboard information is stored locally using browser storage, allowing rankings to persist between visits to the website. Whenever a player's score exceeds their previous best score, the stored leaderboard data is automatically updated. This ensures that the rankings always reflect each user's highest achievement while maintaining accurate and up-to-date results.

The ranking system encourages repeated gameplay and continuous improvement by motivating players to achieve higher scores and improve their position on the leaderboard.



Figure 9: Leaderboard showing the highest-ranked players

Profile

The profile page displays information specific to the logged-in user. This includes personal details such as username and email address, together with gameplay statistics.

Displayed statistics include total games played, highest score achieved, best words-per-minute result, best accuracy percentage, and levels unlocked.

The profile updates automatically after each completed game session.

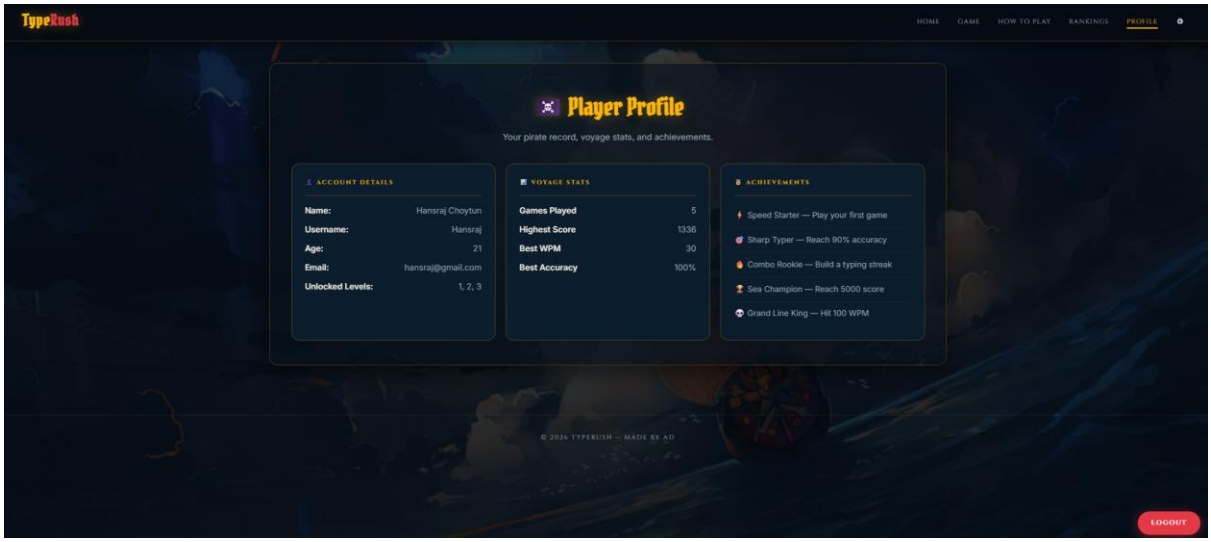


Figure 10: Profile page when user logged in

Settings

The settings section allows users to customise aspects of the game experience.

Users can enable or disable sound effects and background audio. Brightness controls are also provided to improve visibility and accessibility depending on user preference.

Links to the Terms and Conditions and Privacy Policy are available through the settings interface.

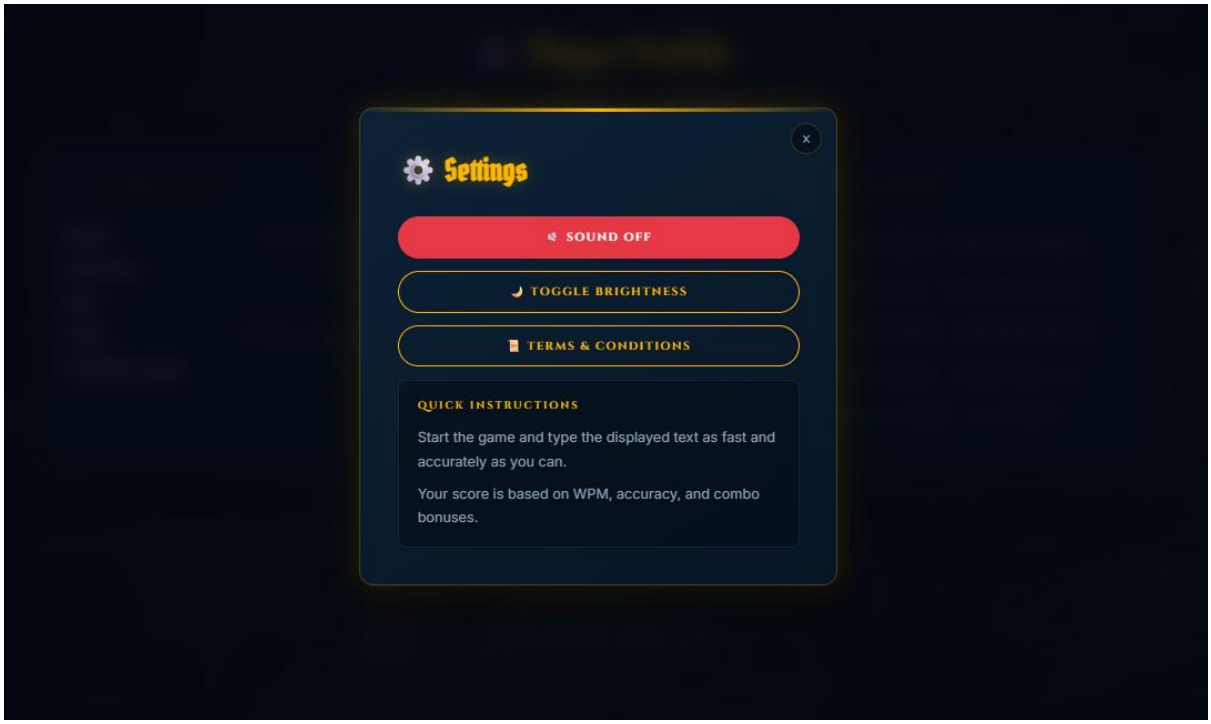


Figure 11: Settings Modal

Legal Pages

Terms and Conditions

The Terms and Conditions page outlines acceptable use of the website, user responsibilities, account requirements, and general rules for accessing the game.

Privacy Policy

The Privacy Policy explains what user information is stored locally, how it is used within the website, and how user privacy is maintained.

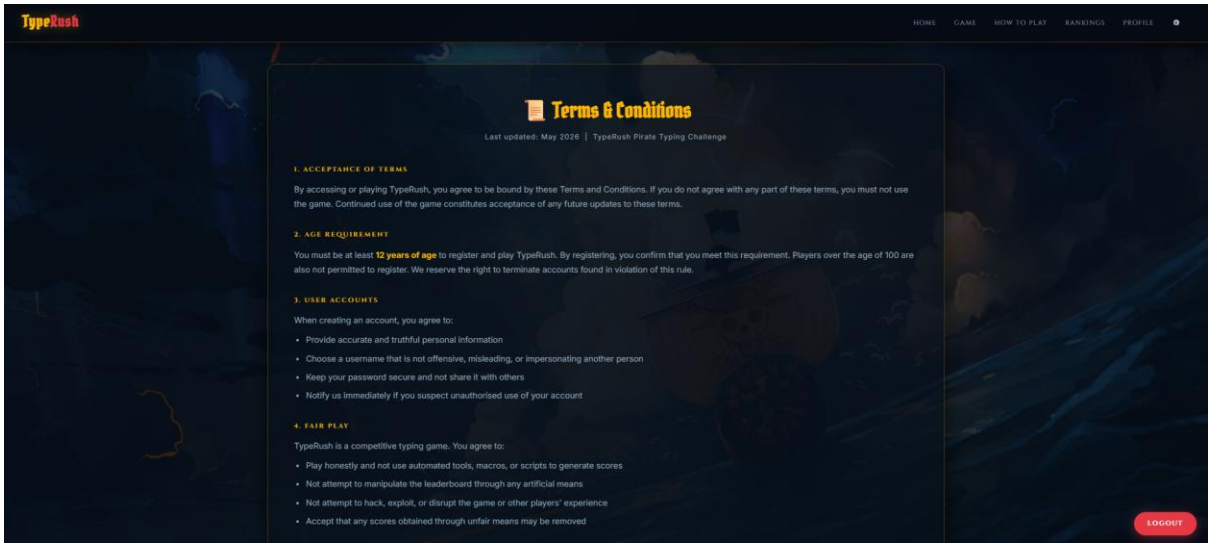


Figure 12: Terms and Conditions Page

Data Storage

TypeRush uses browser-based storage to maintain user information and game progress. User accounts, game statistics, leaderboard information, and unlocked levels are stored locally within the user's browser, allowing progress to persist between visits.

Temporary settings and session information may also be stored during active use of the website. This approach provides a simple method of maintaining functionality without requiring a dedicated server or external database.

Using browser-based storage also improves the user experience by allowing players to continue from where they left off without needing to create a new account each time they visit the website. Personal statistics, high scores, leaderboard positions, and unlocked levels can be retrieved automatically when the user logs in. This ensures that progress is retained while keeping the implementation simple, lightweight, and suitable for a client-side web application.

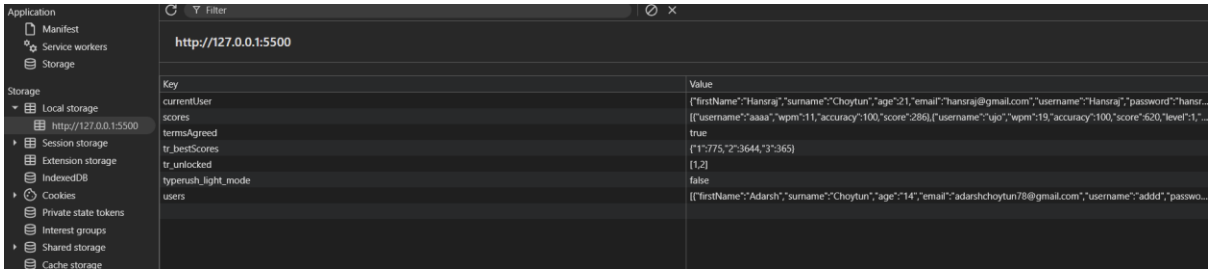


Figure 13: Local Storage Data stored

Development Challenges

Challenge 1 – Level Progression

One challenge involved implementing level unlocking while ensuring that player progress was stored correctly. This was solved by creating user-specific progress tracking and updating unlocked levels after successful completion of each challenge.

Challenge 2 – Live Validation

Implementing registration validation required ensuring that usernames, email addresses, and passwords were checked in real time. Several validation issues were resolved by refining event listeners and feedback messages.

Challenge 3 – Audio Dictation

The audio system required additional debugging to ensure recordings could be played, paused, and replayed correctly across different browsers. This was solved through improved audio state management and user controls.

Reflection and Potential Improvements

The development of TypeRush provided valuable practical experience in web development, user interface design, browser storage, and JavaScript programming. Throughout the project, I gained a deeper understanding of how HTML, CSS, and JavaScript work together to create a complete interactive web application.

Several challenges were encountered during development. One of the main difficulties was implementing the user account system while ensuring that registration validation, login authentication, and user data storage worked correctly. Additional challenges included developing the level progression system, managing leaderboard updates without creating

duplicate entries, implementing audio controls for the dictation level, and ensuring that player progress and statistics were stored correctly using browser storage. These issues were resolved through testing, debugging, and incremental improvements to the code.

The project also provided experience in organising code using modules and classes, improving code readability and maintainability. Careful attention was given to usability, navigation, visual design, and error handling to create a more polished user experience.

If additional development time were available, several improvements could be implemented. These include online accounts with cloud storage, multiplayer functionality, additional levels, more audio challenges, achievement systems, and a database-backed leaderboard that would allow users to compete globally.

Overall, the project successfully achieved its objective of creating an engaging typing game while demonstrating the use of registration, login, validation, local storage, rankings, and interactive gameplay. The development process provided valuable experience in software design, problem solving, debugging, and web application development.

Use of AI

Artificial Intelligence tools, specifically ChatGPT and Claude Ai, were used during the development of this project as a learning and support resource. AI was used to explain programming concepts, suggest possible solutions to coding problems, assist with debugging, and provide feedback on website design and structure.

The tool was also used to clarify JavaScript, HTML, and CSS concepts when difficulties were encountered during development. However, all final decisions regarding design, implementation, testing, and functionality were made by the developer. Any code generated with AI assistance was reviewed, modified where necessary, and integrated only after it was fully understood.

AI was therefore used as a learning aid and productivity tool rather than as a replacement for independent development.

References

Canva (2026) *Canva Graphic Design Platform*. Available at: https://www.canva.com (Accessed: June 2026).

GeeksforGeeks (n.d.) *W3Schools*. Available at: https://www.geeksforgeeks.org/W3Schools (Accessed: June 2026).

Gamez, D. (2025) *CST2120 Web Applications and Databases Video Playlist*. YouTube. Available at: https://www.youtube.com/watch?v=MvgjaY_m5E0&list=PLQbUxXl36NPju2zCrphcoki5XqUIGPWZU (Accessed: June 2026).

JavaScript.info (n.d.) *The Modern JavaScript Tutorial*. Available at: https://javascript.info (Accessed: June 2026).

MDN Web Docs (n.d.) *Web Development Documentation*. Available at: https://developer.mozilla.org (Accessed: June 2026).

OpenAI (2026) *ChatGPT*. Available at: https://chatgpt.com (Accessed: June 2026).

W3Schools (n.d.) *HTML, CSS and JavaScript Tutorials*. Available at: https://www.w3schools.com (Accessed: June 2026).

Mozilla Developer Network (n.d.) *JavaScript Modules*. Available at: https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide/Modules (Accessed: June 2026).

Google Fonts (n.d.) Available at: https://fonts.google.com (Accessed: June 2026).